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# Living Research Manuscript

**Team Name:**

*Team Retail Electricity Prices*

**Team Members:**

*Vijay Hong, Tarun Tammali, Claire Olotara*

**Project Title:**

*Data Center Load & Virginia Residential Electricity Prices (2018-2025)*

**Abstract**

Rapid data center expansion has increased concerns about rising electricity demand, yet its impact on residential retail electricity prices remains unstudied. This study examines the relationship between electricity demand associated with rapid data center expansion and residential retail electricity prices in the global hub for data center development, Virginia, from 2018 to 2025. Specifically, we investigate whether the relationship between residential retail electricity prices, Dominion Energy Territory (DOM) electricity demand, and natural gas prices has changed following the rapid expansion of Artificial Intelligence (AI) beginning in late 2022.

Using data from the U.S. Energy Information Administration (EIA), we employ time-series analysis to estimate the relationship between residential prices, natural gas prices, and DOM electricity demand, controlling for weather, customer counts, and generation mix. We also use Geographic Information System (GIS) maps from ArcGIS and the Cleanview Data Center Tracker to visualize the spatial distribution and concentration of data centers.

Results show that residential prices track natural gas costs at a roughly six-month lag, consistent with the regulated fuel factor and rate case timing. The association between DOM load and residential prices, by contrast, is highly sensitive to the secular time trend and cannot be cleanly separated from it in this time series design. We find no detectable improvement in the association following the AI-driven expansion. Two structural breaks in the price series align with the natural gas price cycle, not with data center growth. The GIS map confirms that Virginia's data centers are concentrated in Loudoun and Prince William counties within the DOM zone. These findings motivate the multistate panel design and can inform utility planning, infrastructure investment, electricity affordability, and regulatory decision-making as data center expansion continues.

**Keywords:** retail electricity; residential electricity prices; data center; artificial intelligence; Virginia; GIS mapping; time-series analysis; natural gas prices; Dominion Energy; electricity demand

## 1. Introduction

With the rapid advancements in technology and the expansion of the Internet of Things (IoT), the demand for data centers has grown significantly to support the increased volume of digital data worldwide (Koot & Wijnhoven, 2021). This demand is accelerated even further with the late 2022 artificial intelligence (AI) boom, as the widespread use of generative artificial intelligence and other computationally heavy applications requires substantial computing infrastructure. As a result, the number and size of data centers have expanded rapidly, leading to an increase in electricity demand. This growing electricity demand raises concerns about the ability to meet future electricity needs while maintaining affordable and reliable service.

Recent literature has shown that the growing electricity demand from data centers has been a crucial factor in US electricity markets. Studies show that retail electricity prices vary widely across states due to regional differences in power infrastructure, energy generation, and utility policies (Kay et al., 2026; Wiser et al., 2025). Evidence suggests that this creates an overlooked pressure on computational performance and power grids that require greater electricity demand. (Chen et al., 2025). However, electricity supply is limited, suggesting that increases in electricity usage, technological upgrades, data center operations, and grid upgrades will drive higher power generation, further affecting retail electricity prices. (Makholm & Olive, 2024; Xu & Li, 2014). Moreover, inelastic residential demand can result in greater cost pass-through, correlating with residtianl customer more directly (Addey et al., 2019). Together, these findings suggest that regional factors play a key role in the relationship between data center growth and retail electricity prices.

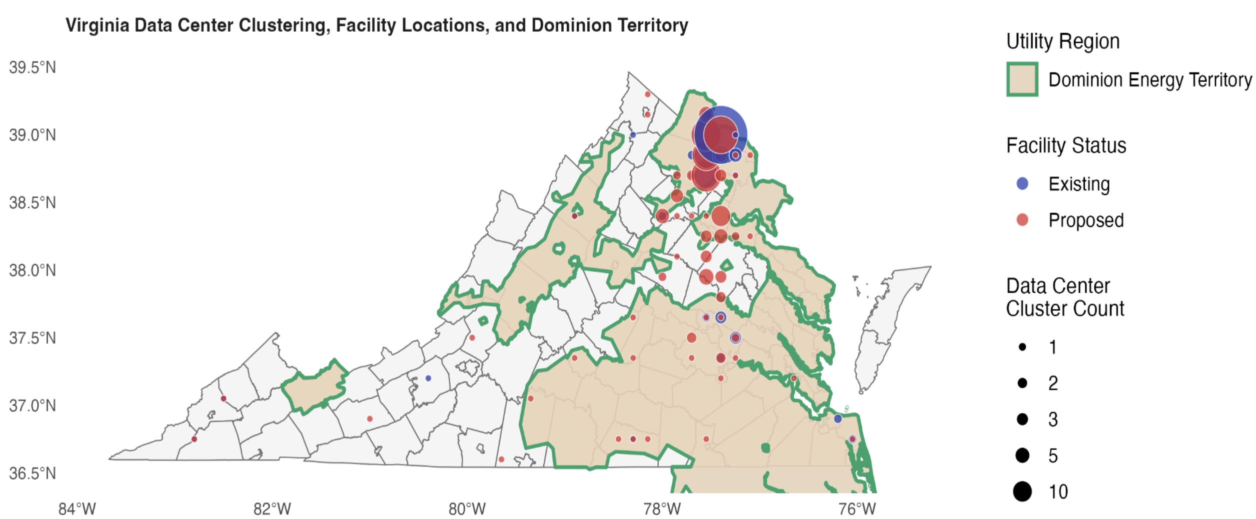


**Figure 1: Data Center Clustering in Virginia**

*Note. This figure illustrates the geographic distribution of existing and proposed data centers across Virginia within the Dominion Energy Territory (DOM), the largest PJM zone in the state. It reveals a strong spatial concentration of facilities in Northern Virginia and the DOM territory, especially in Loudoun County and the Washington, D.C. region. The area has the highest density of data centers, as indicated by the* *large red bubble, while the rest of the state has few facilities that are more widely spaced. This motivates our study analyzing the association between residential retail electricity prices and localized data center expansion in DOM, specifically their electricity load.*

Our paper examines and provides empirical evidence for the influence of data center expansion on residential retail electricity prices in Virginia. We contribute to prior literature (e.g., Kay et al., 2026; Feher et al., 2025) on the influence of data center expansion on residential retail prices before the late 2022 AI boom. We also provide new evidence on how their relationship has evolved in recent years following the AI boom, and on how residential retail prices are projected to evolve by observing current AI-driven electricity market trends. Overall, our findings provide evidence that may aid policymakers, utility companies, or regulators in making crucial decisions regarding infrastructure planning and cost allocation, grid planning, electricity affordability and market regulation, and future data center electricity demand.

**Research Questions**

1. Do the electricity demands of Virginia's data centers show up in residential electricity bills?
2. Has this relationship changed since the late-2022 AI boom?

**Objectives**

- Investigate the relationship between residential electricity prices and Dominion Energy Territory (DOM) load.
- Investigate the relationship between residential electricity prices and natural gas prices.
- Observe whether both relationships with residential electricity prices change after the late 2022 AI boom.

## 2. Data

Our main source of average sector prices is EIA-861 (US Energy Information Administration), which reports residential-sector revenue, prices, sales, and customers from 2018 to 2025. The dataset includes all US states, but for our analysis, we use only Virginia. Similarly, yearly Virginia electricity consumption per capita was taken from the EIA. The dataset spans 1960 to 2025, with our analysis focusing on 2018 to 2025. The DOM datacenter load was obtained from the PJM Data Miner.

Our choice to model electricity demand in the DOM transmission PJM zone (GWh), is driven by the uniquely accelerating load growth and the number of data centers in Virginia being primarily located in DOM territory. As shown in Figure 2, the ratio of DOM load to total Virginia retail sales shows a positive slope over the sample period, accelerating rapidly after 2022. The figure also shows DOM load tracks Virginia's total retail electricity sales accurately, with a Pearson correlation coefficient of 0.97 in levels and 0.95 in first differences.

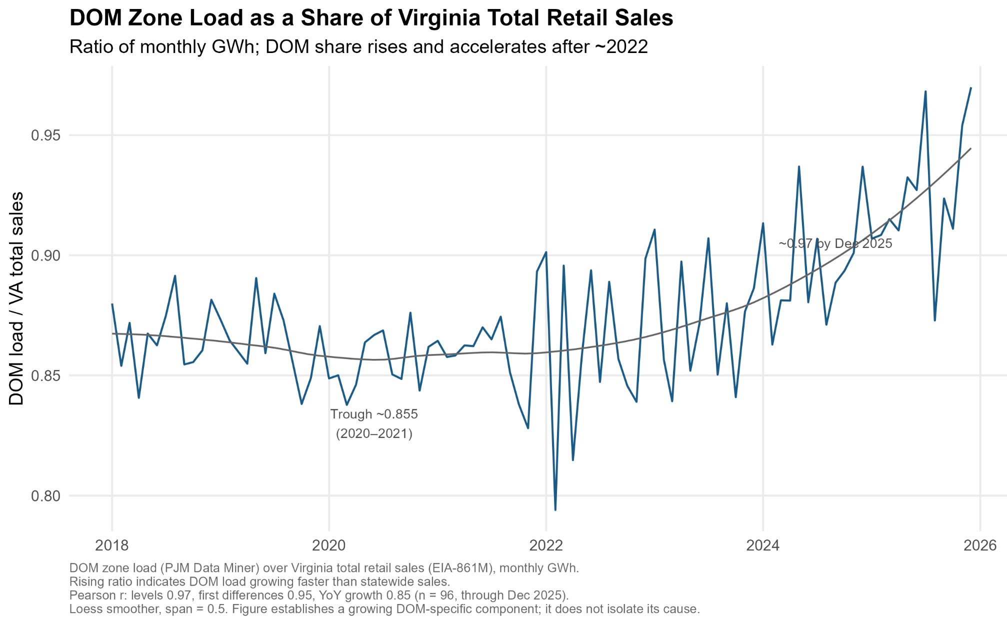


**Figure 2: DOM Load Share of Virginia Retail Sales (2018-2025)**

*Note. This figure shows the relationship between DOM Zone Load and Virginia Retail Sales. The primary finding is that the DOM zone's share of Virginia's total electricity sales has increased since around 2022, suggesting that electricity demand in the DOM region has grown at a rapid rate. This motivates our study analyzing the association between residential retail electricity prices and localized data center expansion in DOM, specifically their electricity load.*

We obtain precise Virginia data center locations from ArcGIS and the Cleanview data center tracker. The key variables we use are building size, reported megawatts, and precise location, spanning 2024 to 2026. For our control variables, we used the yearly Virginia population data from the Federal Reserve Bank of St. Louis, covering 2018-2025, with estimates for 2026. In addition, the US yearly inflation rate was taken from the Federal Reserve Bank of Minneapolis and covers 2018 to 2025.

**Table 1: Summary Statistics**

| Dataset Summary Statistics | | | | | | |
|---|---|---|---|---|---|---|
| **Variable** | **n** | **Mean** | **Median** | **SD** | **Min** | **Max** |
| Residential Electricity Price | 96 | 13.19 | 12.61 | 1.44 | 10.88 | 16.62 |
| Natural Gas Prices | 96 | 17.69 | 17.24 | 5.11 | 9.41 | 28.44 |
| Heating Degree Days | 96 | 160.0 | 100.8 | 163.4 | 0.0 | 536.9 |
| Cooling Degree Days | 96 | 79.4 | 25.9 | 96.5 | 0.0 | 311.7 |
| Average Load DOM | 96 | 12,831.5 | 12,762.2 | 1,858.7 | 9,375.0 | 17,994.6 |
| Revenue | 96 | 505,168.7 | 491,044.5 | 109,882.4 | 300,051.0 | 832,754.0 |
| Sales | 96 | 3,850,573.6 | 3,787,676.5 | 827,130.0 | 2,417,739.0 | 5,997,961.0 |
| Customers | 96 | 3,561,873.4 | 3,569,980.5 | 85,947.7 | 3,419,165.0 | 3,707,509.0 |
| Total Generation | 96 | 7,837.1 | 7,695.4 | 1,325.3 | 4,998.5 | 11,193.7 |

*Note: Table 1 presents the summary statistics for our primary variables of interest in this study. It includes 96 monthly observations spanning 2018 to 2025. For each variable, the table reports the number of observations, mean, median, standard deviation (SD), minimum, and maximum values.*

The table shows that residential retail electricity prices averaged 13.19 cents per kWh, with values ranging from 10.88 to 16.62 cents per kWh. The average electricity load for DOM was 12,831.5 GWh, while the average natural gas price was 17.69. Weather conditions varied, with Heating Degree Days (HDD) averaging 160 while Cooling Degree Days (CDD) averaged 79.4 per month. Other variables, such as revenue, electricity sales, customer count, and total electricity generation, also show considerable variation across the study period.

## 3. Methods

We use time-series regression to estimate the relationship between the DOM zone and electricity load, natural gas prices, and Virginia residential retail electricity prices from 2018 to 2025. Prices are deflated to real terms using the Consumer Price Index, so that price movements reflect changes in electricity costs rather than general inflation. The model controls for weather(heating and cooling degree days), residential customer counts, natural gas generation share, and seasonality through month fixed effects.

Because electricity prices, load, and gas prices trend over the period selected, we first test each series for stationarity using the Augmented Dickey-Fuller and KPSS tests. The series contains unit roots in levels and is stationary in differences, indicating shared trends. This has a direct consequence for interpretation, for when both the dependent variable and a regressor trend together, their association in levels may partly reflect a common trend rather than a genuine relationship. We therefore estimate the model in both levels and first differences and assess the stability of each coefficient across specifications rather than relying on a single estimate.

The levels specification is:

$$P_t = \beta_0 + \beta_1\ G_{t-6} + \beta_2\, L_t + \beta_3\, B1_t + \beta_4\, B2_t + \beta_5\, (L_t \times Post\text{-}AI) + \delta_m\, mon_{m(t)} + \lambda\, trend_t + \varepsilon_t \qquad (1)$$

where $P$□ is the real Virginia residential electricity price in month t, $G_{t-6}$ is the real Henry Hub natural gas price lagged 6 months, and $L_t$ is the DOM zone electricity load. $\beta_1$ is the gas coefficient indicator, and $\beta_2$ is the transmission load, and it gives the load-price association in the pre-period. $\beta_3$ and $\beta_4$ are structural break indicators for October 2021 and March 2023, located empirically using a supF test. $\beta_5$ is the coefficient on the interaction between DOM load and the post-2022 indicator, and it captures whether the association between load and real price differs after the AI-driven expansion. The term $mon_{m(t)}$ represents the month fixed effects, $trend_t$ is the linear time trend, and $\varepsilon_t$ is the unexplained variation. All standard errors are Newey-West (HAC), robust to autocorrelation and heteroskedasticity. In other words, it reads as: this month's real price equals a baseline, plus the association with gas prices six months ago, plus the association with contemporaneous load, plus a level shift at each break, plus seasonal and trend controls, plus noise.

We estimate this model as a sequence of specifications that differ only in how co-trending controls are treated, progressively replacing the linear trend with the dated break indicators and adding or removing customer counts. This specification ladder tests whether the load price association is robust to how the shared trend is handled.

As a further check against the non-stationarity identified above, we estimate a first-difference specificatio

$$\Delta\ Pt = \beta 0 + \beta 1\ \Delta\ Gt-6 + \beta 2\ \Delta\ Lt + \varepsilon t \qquad (2)$$

Where Δ denotes the monthly change in each variable, differencing removes the shared trend. Hence, a coefficient that is stable in differences reflects a short-run association rather than a common trend.

We also use GIS mapping to visualize the geographic distribution and concentration of data centers across Virginia, overlaying facility locations on county boundaries and the Dominion service territory. This descriptive map identifies areas of high data center concentration and supports using the DOM zone as a proxy for Virginia demand.

## 4. Results

Table 2 presents the main results from the time-series regression models examining the relationship between Virginia residential retail electricity prices and the explanatory variables. The DOM load coefficient is positive and significant in the trend-controlled and break-controlled specifications (M2, M3: 0.001, p<0.001) but marginal when customer counts are removed (M4) and not different from zero in the first differences (M5). When we add the break-controlled model with a post 2022 interaction, the interaction term is not significant (M6), indicating no detectable change in the load-price association after the AI-driven expansion. Unit root tests indicate load and price are non-stationary in levels and stationary in differences. The six-month lagged real Henry Hub natural gas prices are the most consistent predictors of Virginia's residential electricity prices. It is positive, statistically significant, and consistent in Models 1 through 6, with coefficients ranging from 0.133 to 0.193 ($p < 0.05$), whereas the three- and twelve-month lags are insignificant. DOM load is positive and significant in Models 2 and 3 with a coefficient of 0.001 ($p < 0.001$), but loses significance in Models 1, 4, and 5, indicating that its association is sensitive to the specifications. In the first-difference model (M5), lagged natural gas prices remain significant, indicating that they are a robust driver of residential electricity prices. Model 6 introduces a post-2022 AI interaction to test whether the relationship between DOM load and residential electricity prices changed after the AI boom. Similarly, the post-AI indicator is positive and statistically significant at 1.816 ($p < 0.05$); the interaction between the DOM load and post-AI is not statistically significant, suggesting that the association between electricity load and electricity prices did not strengthen after the AI boom. Overall, the results show that the relationship between residential electricity prices and lagged natural gas is consistent.

## Table 2: Time Series Analysis Results

Real Virginia residential price: specification ladder and post-AI test (HAC SE)

| | M1: +trend | M2: -trend | M3: +breaks | M4: -customers | M5: 1st diff | M6: interaction |
|---|---|---|---|---|---|---|
| DOM load (GWh) | 0.001+ | 0.001*** | 0.001*** | 0.000+ | 0.000 | 0.001*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Post-AI (2023+) | | | | | | 1.816* |
| | | | | | | (0.703) |
| DOM load x Post-AI | | | | | | -0.000+ |
| | | | | | | (0.000) |
| Real gas, 6-mo lag | 0.137** | 0.141** | 0.172*** | 0.193*** | 0.142** | 0.133*** |
| | (0.046) | (0.049) | (0.036) | (0.050) | (0.053) | (0.037) |
| Real gas, 3-mo lag | -0.036 | -0.055 | 0.077 | 0.094 | -0.005 | 0.084 |
| | (0.035) | (0.044) | (0.060) | (0.072) | (0.059) | (0.057) |
| Real gas, 12-mo lag | -0.046+ | -0.037 | -0.012 | 0.011 | 0.007 | -0.013 |
| | (0.026) | (0.031) | (0.026) | (0.032) | (0.037) | (0.028) |
| Gas generation share | 6.943* | 10.351*** | 11.935*** | 8.807** | 4.353* | 11.824*** |
| | (3.264) | (2.889) | (2.301) | (2.948) | (1.768) | (2.231) |
| Residential customers (000s) | -0.046** | -0.018*** | -0.011*** | | -0.014* | -0.013*** |
| | (0.014) | (0.003) | (0.003) | | (0.006) | (0.003) |
| Break: Oct 2021 | | | -1.222*** | -1.625*** | | -1.190*** |
| | | | (0.287) | (0.324) | | (0.269) |
| Break: Mar 2023 | | | 0.635* | 0.761* | | |
| | | | (0.283) | (0.345) | | |
| Linear trend | 0.112* | | | | | |
| | (0.055) | | | | | |
| Num.Obs. | 84 | 84 | 84 | 84 | 83 | 84 |
| R2 | 0.692 | 0.652 | 0.750 | 0.693 | 0.511 | 0.767 |
| R2 Adj. | 0.595 | 0.548 | 0.666 | 0.596 | 0.364 | 0.683 |
| Std.Errors | Custom | Custom | Custom | Custom | Custom | Custom |

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Newey-West HAC standard errors. Month dummies included, not shown.
M5 is in first differences. M6 adds the post-AI interaction (Objective 2 test).
Coefficients are conditional associations. Load moves across M1-M5;
the 6-month gas lag is stable throughout; the M6 interaction is not significant.

*Note: This regression table presents findings on real residential electricity prices in Virginia. M1 includes a linear time trend; M2 removes the trend; M3 replaces trends with structural break indicators for October 2021 and March 2023; M4 removes residential customer count; and M5 estimates the model in first differences; M6 adds an interaction between DOM load and post-2022 indicator. Results show that the six-month natural gas price lag is a dominant, robust driver, and DOM load is sensitive to the model specifications.*

Figure 3 displays the estimated lag of natural gas pass-through to Virginia's residential retail electricity prices. Figure 3 shows a non-linear peak in the natural gas pass-through at six months. At the 6-month lag, the coefficient is statistically significant and positive, with a point estimate near 0.17 and a 95% confidence interval above 0. In contrast, the estimated coefficients for both the 3-month and 12-month lags are not statistically significant at the 95% confidence level. This pattern reveals that fluctuations in natural gas prices and residential retail electricity prices are not immediate; rather, they operate in 6 months.

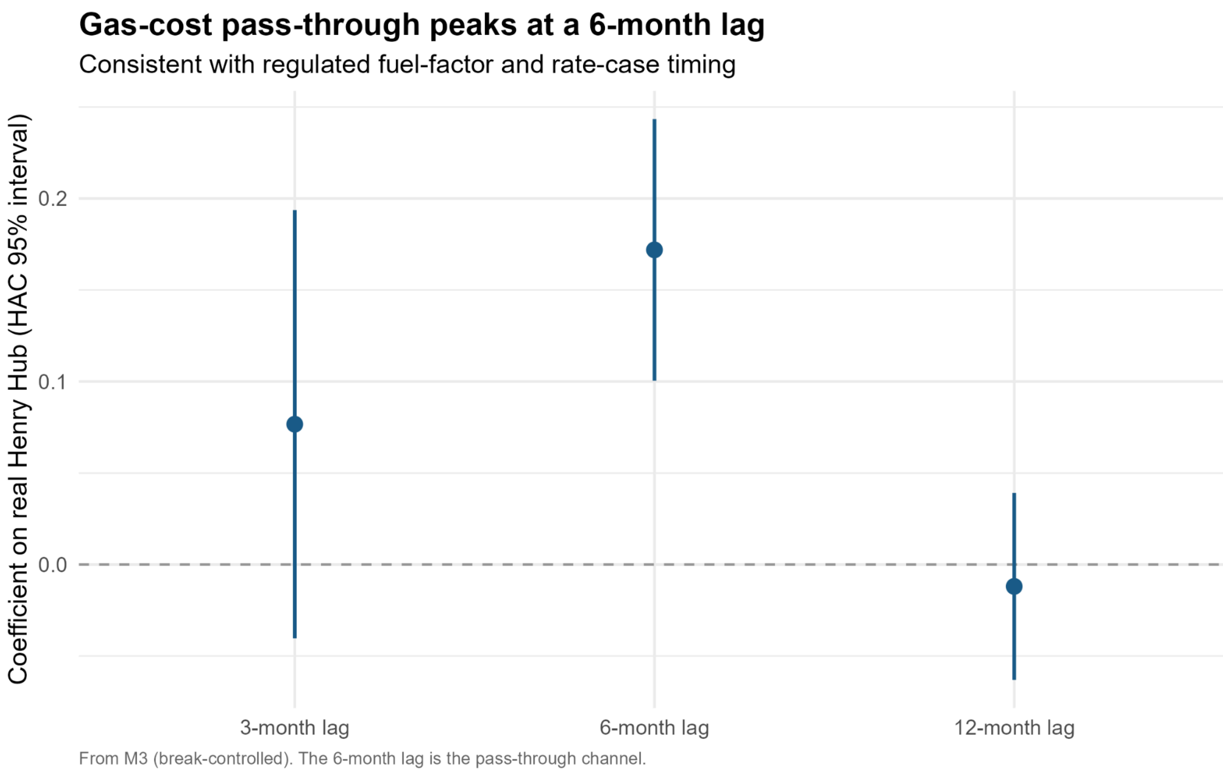


**Figure 3: Gas-Cost Pass-Through Coefficients by Lag Duration**

*Note: This figure illustrates the relationship between gas costs and electricity prices across three different time lags. It shows that the association is significant and peaks at a 6-month lag, while the 3-month and 12-month intervals are less consistent.*

Figure 4 presents a coefficient stability analysis across four model specifications (M1 to M4). This comparison answers whether the relationship between residential electricity prices and our primary drivers is robust to changes in secular controls. The comparison demonstrates that the 6-month lagged natural gas fuel pass-through remains robustly positive and significant. In contrast, the statistical significance of the load coefficient is highly sensitive to the inclusion or exclusion of secular controls.

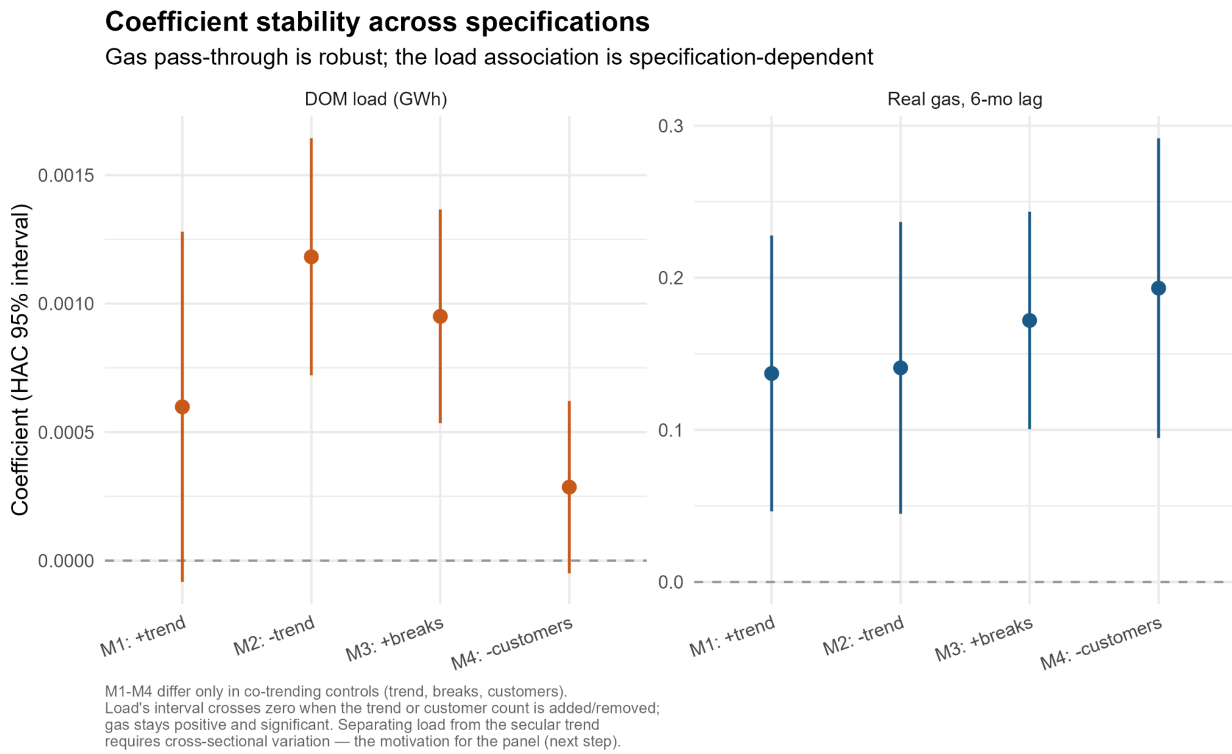


**Figure 4: Coefficient Stability Across Specifications**

*Note: This figure shows the coefficient stability across alternative models. Point estimates and 95% HAC confidence intervals compare DOM electricity load (GWh) against a 6-month lag of real natural gas prices across models M1 to M4.*

## 5. Discussion

This study asked whether data center expansion increases residential retail electricity prices in Virginia and whether that relationship evolved after the 2022 AI boom. Overall, our findings provide little evidence that statewide electricity demand powered by DOM load is a strong driver of residential electricity prices. Instead, lagged natural gas prices emerged as the consistent predictor of residential electricity prices, revealing that fuel costs play a larger role in explaining them.

While Wiser et al.'s 2025 study establishes that state exposure to natural gas price risk increases retail electricity prices, our findings further extend this framework by quantifying the lagged pass-through under regulated-rate settings. As shown in Figure 3, natural gas prices at a 6-month lag show a positive and statistically significant relationship with inflation-adjusted residential electricity prices. This lagged pass-through is consistent with Virginia's regulated rate-setting: fuel-factor adjustments and State Corporation Commission (SCC) rate cases transmit wholesale cost changes to residential bills over roughly two quarters rather than immediately. This lagged structural delay aligns with Hausman (2019), who notes that public utility frameworks inherently use bill-smoothing adjustments that insulate consumers from immediate wholesale shocks. These empirical findings have practical implications for policymakers and key stakeholders in

Virginia's electricity market. The robust, 6-month lag in natural gas price passthrough confirms that existing regulatory smoothing mechanisms are working as designed. However, because this rate process operates with a lag, regulators must anticipate that residential prices will remain elevated for several quarters even after wholesale gas prices have peaked and started to decline.

The coefficient stability analysis shown in Figure 4 reveals that while the 6-month lagged natural gas price is a robust driver of retail electricity prices, the association with DOM load is highly fragile and dependent on the controls. When accounting for linear time trends or regional customer growth, the statistical significance of the DOM load variable disappears, indicating that the time-series data can't isolate data center demand shocks from broader secular trends. The coefficient instability, together with its change in direction and significance under differencing in Table 2, indicates that load growth cannot be empirically separated from the secular time trend in a single series due to collinearity over the sample period. For policymakers focused on cost allocation and infrastructure planning, this shows aggregate statewide metrics are not as reliable. To protect residential consumers from unfair costs, policymakers must demand more localized utility data and spatial analysis to understand how localized transmission and generation upgrades are being funded. Our findings mirror Millican et al's 2026 study, which shows that rising retail rates are driven by baseline system trends and structural utility capital expenditures rather than demand growth itself. Resolving this requires cross-sectional comparison, which motivates a panel data design for future steps.

**6. Conclusion**

This study examines the relationship between the rapid expansion of data centers and residential retail electricity prices in Virginia. As advancements in artificial intelligence and cloud computing continue to increase demand for data centers, understanding their impact on electricity markets has become increasingly important. The primary objective of this research was to determine whether data center expansion is associated with changes in residential electricity prices while accounting for other factors that influence electricity costs, such as electricity generation, natural gas prices, and temperature. Empirical findings show residential price fluctuations are robustly associated with 6-month lagged natural gas prices, while the correlation with DOM load is highly sensitive to secular time trends. Along with evidence on residential retail electricity prices and data center expansion, this study provides an initial framework for a panel data design to evaluate how increasing data center electricity demand may affect retail electricity prices and to identify opportunities for future research through forecasting and more detailed spatial analyses.

Several important questions remain unanswered, providing opportunities for future research. One limitation of this study is the reliance on state-level average electricity prices, which obscures county-level trends. As a result, we are unable to directly compare residential retail prices with GIS-mapped data center locations to determine whether areas with a higher concentration of data centers exhibit different price trends than surrounding regions. Incorporating county-level electricity price data would enable a more detailed spatial analysis and provide greater insight into the localized impacts of data center development. Additionally, our analysis relies on a single state-level retail electricity price time series, which cannot fully separate the effects of increasing data center electricity demand from broader secular trends in electricity prices. A more comprehensive forecasting model requires data from neighboring states because electricity markets are highly interconnected, and changes in one state may influence prices in others through regional spillover effects. Due to time constraints, we were unable to collect and incorporate this additional data. Furthermore, adopting localized integrated learning and framework could help future models localized demand respoinses while maintaining data privacy (Che et al., 2025). Addressing this would improve forecasting accuracy and provide evidence on how continued data center growth affects future electricity markets, allowing policymakers to develop strategies to mitigate and prepare for significant price increases.

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## Appendix A: Supplementary Material

### Table 3: Stationary Tests for Level Specification & First-Difference Model

| | series | adf_p_level | kpss_p_level | adf_p_diff | kpss_p_diff |
|---|---|---|---|---|---|
| 1 | res_price | 0.164 | 0.01 | 0.01 | 0.1 |
| 2 | avg_load | 0.386 | 0.034 | 0.01 | 0.1 |
| 3 | total_gen | 0.216 | 0.061 | 0.01 | 0.1 |
| 4 | Customers | 0.354 | 0.01 | 0.01 | 0.1 |
| 5 | nat_gas | 0.01 | 0.1 | 0.01 | 0.1 |

*Note: This figure displays the stationary tests for the variables. 'Raw' data was non-stationary, motivating a first difference, which came out stationary in both ADF and KPSS tests.*

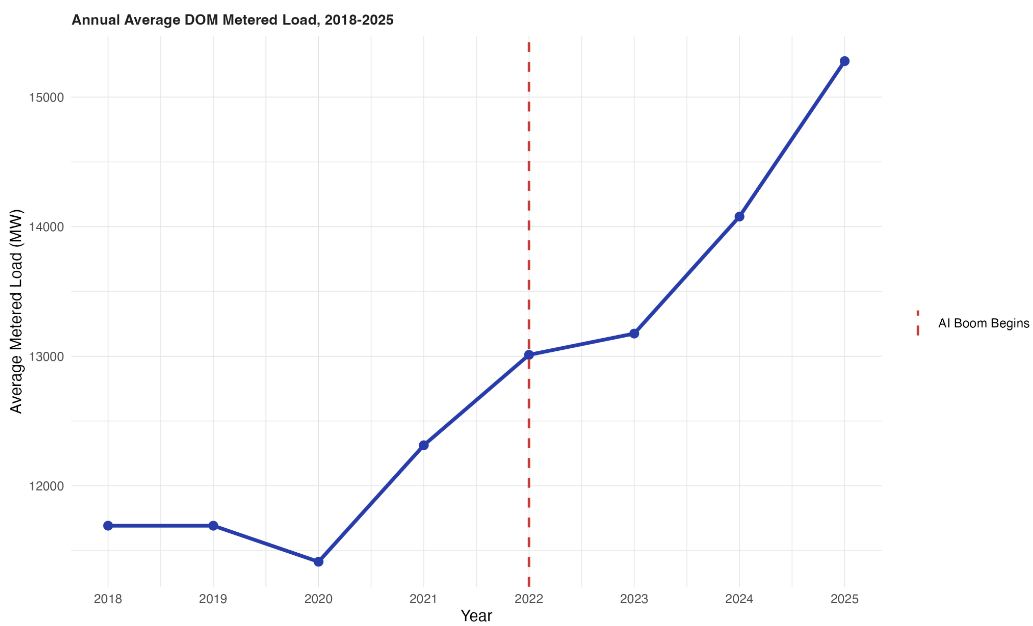


**Figure A1: Annual Average Dominion Energy Metered Load (2018-2025)**

*Note: This figure displays the annual average metered load (MW) for Dominion Energy, illustrating the demand trend over the specified period.*

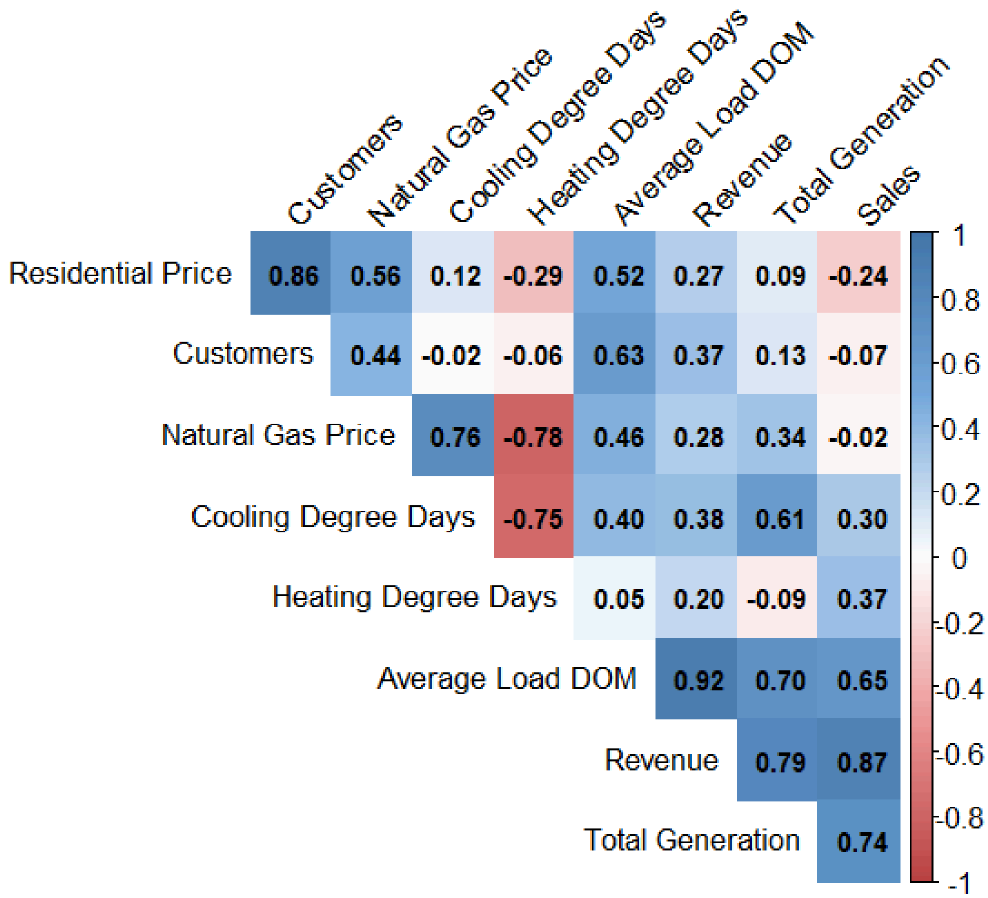


**Figure A2: Correlation Matrix Between Variables**

*Note: This figure shows a correlation matrix among yearly Virginia residential retail electricity price, electricity generation, residential customers, natural gas price, Heating Degree Days (HDD), Cooling Degree Days (CDD), and average datacenter MW per DOM. The plot shows little correlation between residential retail electricity price and electricity generation, residential revenue, HDD, or CDD. Residential retail electricity prices show a moderate correlation with DOM average load and natural gas prices. Residential retail electricity prices show a strong correlation with residential customers.*

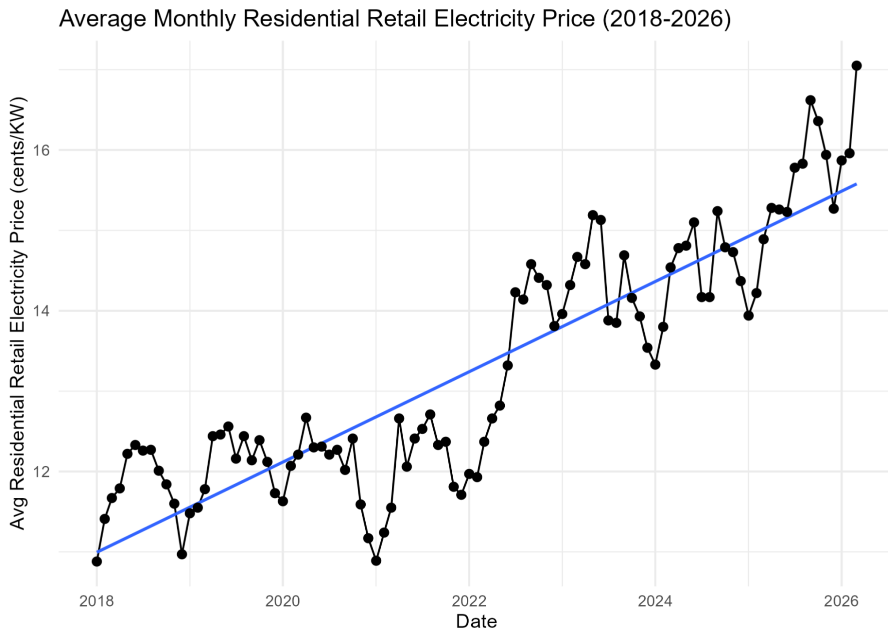


**Figure A3: Trends in Average Monthly Residential Retail Electricity Prices (2018–2026)**

*Note: This figure shows the average monthly residential retail electricity prices, measured in cents/kWh, plotted over time. The graph shows sharp increases and decreases centered around June/July every year, along with sharp spikes in prices from 2022 to mid-2023 and from 2025 to early 2026. Accounting for seasonal fluctuations, there is little increase from 2018 to late 2021. Overall, residential retail electricity prices trend upwards over time.*

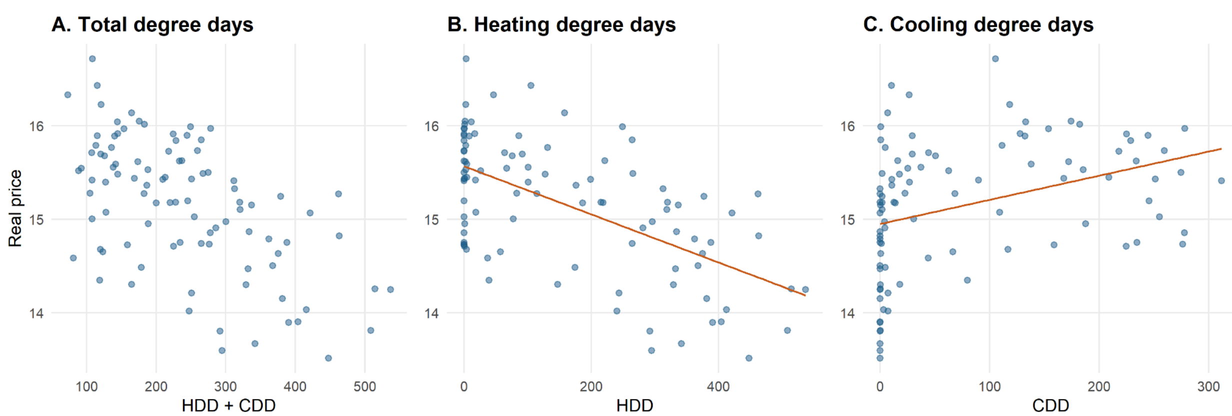


**Figure A4: Relationship Between Real Electricity Prices and Degree Days**

*Note: This figure shows the relationship between residential retail electricity prices and weather demand. Plots show prices plotted against total degree days, heating degree days (HDD), and cooling degree days (CDD). Figures show a positive relationship between prices and CDD, and a negative relationship between prices and HDD.*

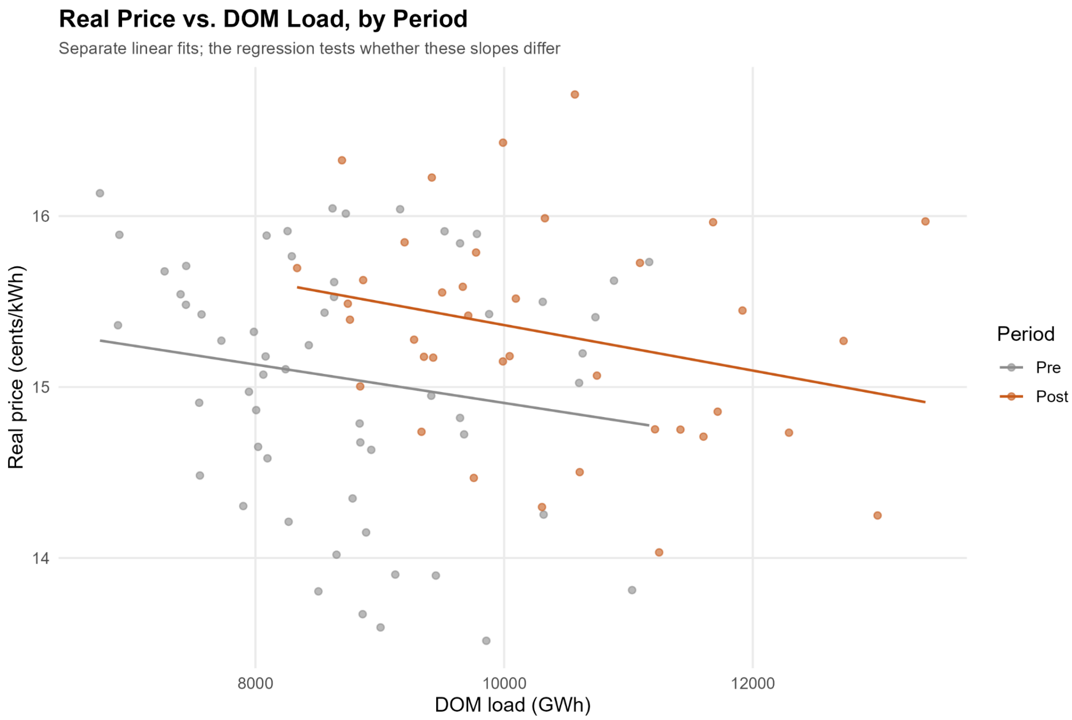


**Figure A5: Relationship Between Real Electricity Price and DOM Load by Period**

*This figure shows real residential electricity prices compared to the DOM load. The plot contains two separate fit lines: pre-AI and post-AI. Results show no significant slope difference between pre- and post-AI.*

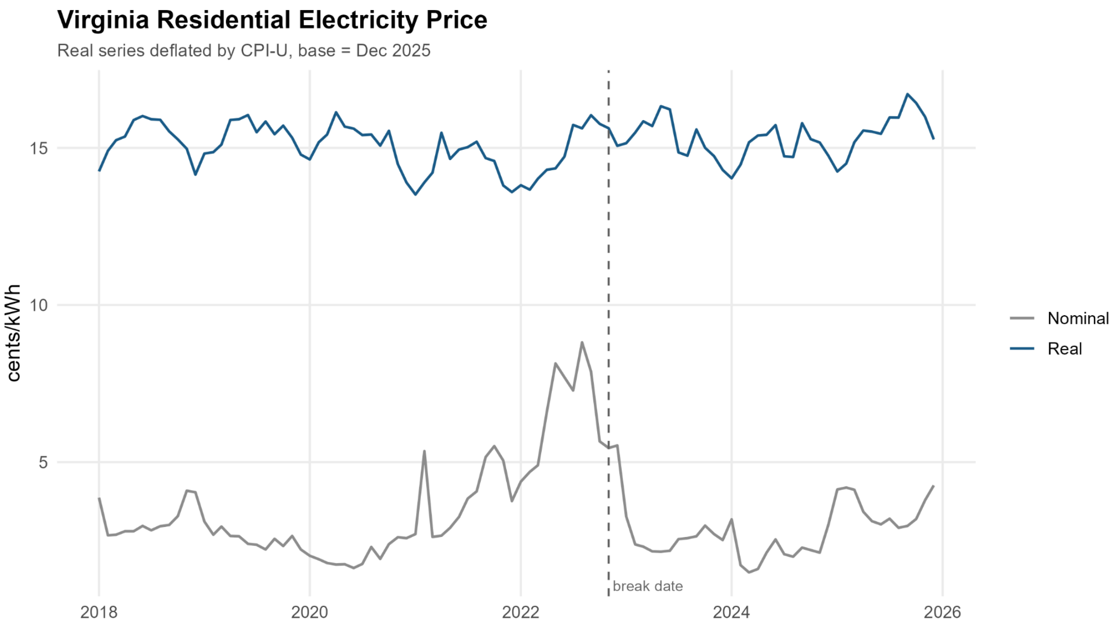


**Figure A6: Time-Series Plot of Nominal and Real Residential Electricity Prices in Virginia**

*Note: This figure compares nominal and real residential electricity prices from 2018 to 2025. The timeline displays nominal rates alongside real prices deflated by CPI-U (base = Dec 2025), with a vertical dashed line tracking a key empirical break date in the series.*

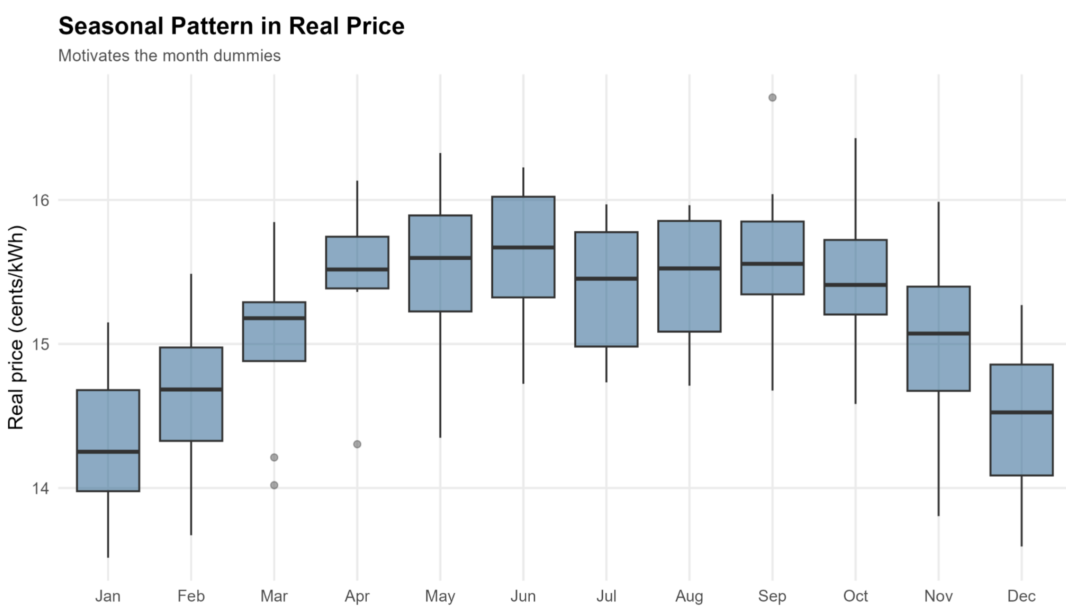


**Figure A7: Seasonal Distribution of Real Residential Electricity Prices in Virginia**

*Note: This figure shows the seasonal distribution of real residential electricity prices by month. Monthly boxplots reveal a non-linear seasonal pattern for prices peaking in May and June, with December and January having the lowest prices.*

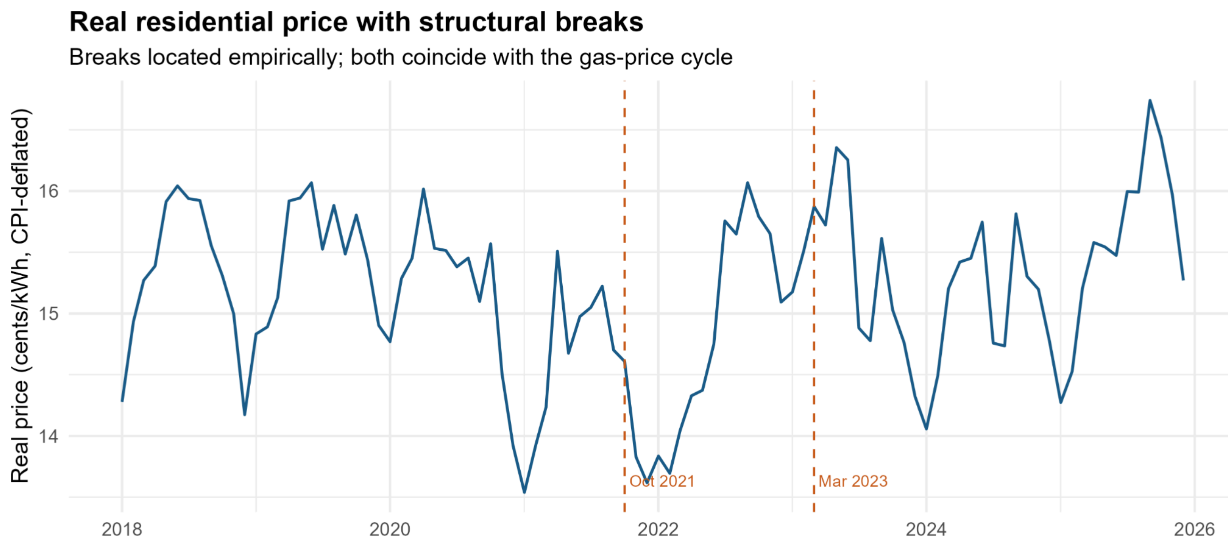


**Figure A8: Real Residential Electricity Prices with Structural Breaks**

*Note. This figure displays Virginia residential electricity prices from 2018 to 2025, adjusted for inflation. The October 2021 break marks the beginning of a decline in real prices, then an increase in 2022, while the March 2023 break indicates a shift to a higher price level.*

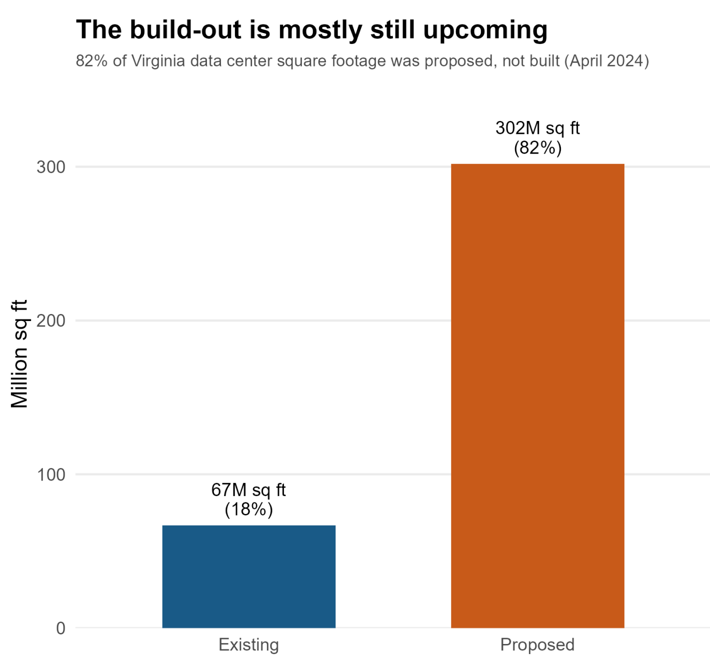


**Figure A9: Comparison of Existing and Proposed Data Center Square Footage**

*Note: This figure shows existing and compared square footage in Virginia. The bar chart compares the existing 67M square feet data centers with the proposed 302M square feet data centers, suggesting that the majority of data center expansion and subsequent impacts is still upcoming.*

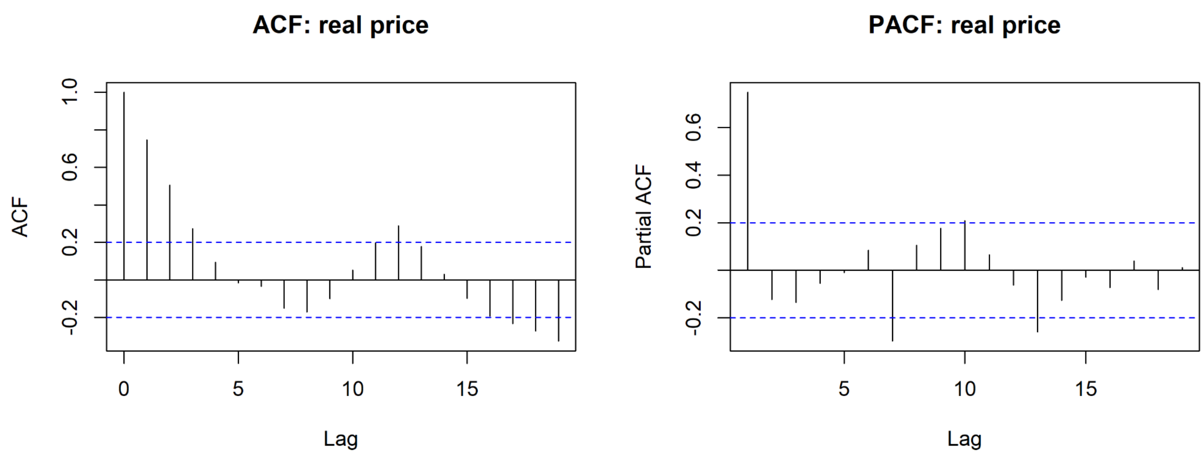


**Figure A10: Autocorrelation and Partial Autocorrelation Functions for Real Electricity Price**

*Note: This figure shows the autocorrelation and partial autocorrelation functions for residential electricity prices. The ACF decays slowly across multiple lags while PCF shows a significant spike at lag 1, indicating strong persistence or non-stationarity in the real price series.*

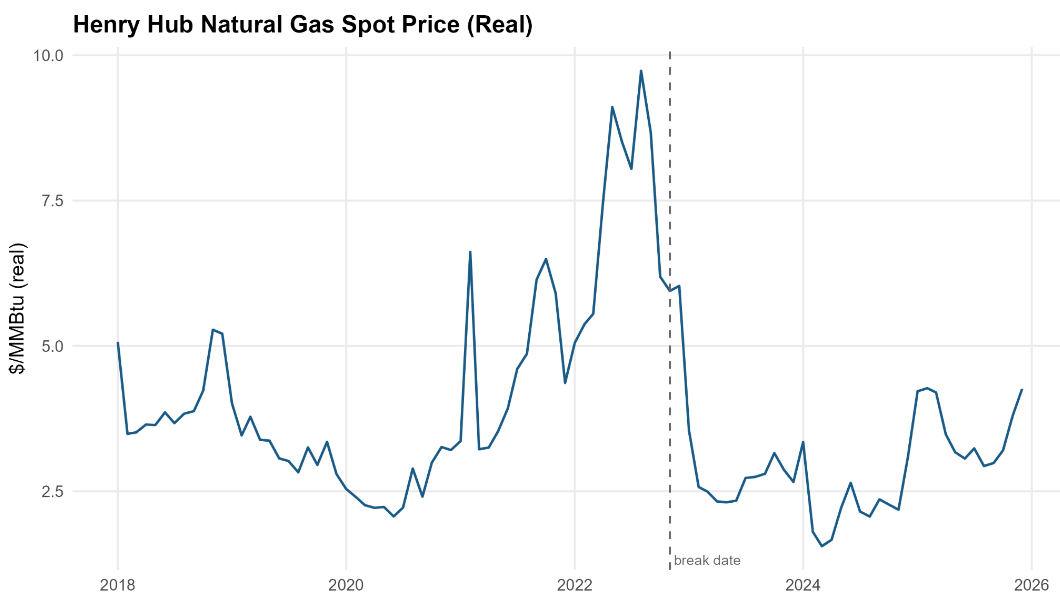


**Figure A11: Adjusted Henry Hub Natural Gas Spot Price**

*Note. This figure shows the trend in inflation-adjusted Henry Hub natural gas spot prices from 2018 to 2025. Natural gas prices remained stable between 2018 and 2021 before rising sharply in 2022, peaking at $10/MMBtu (Million British Thermal Units). Following the structural break, prices declined rapidly, stabilized at lower levels in 2024, and then increased in 2025. It reflects the volatility in fuel costs during the study period and provides context for lagged natural gas prices.*

**Appendix B: Key Variables**

| Variable Name | Role (DV/Control) | Source | Geographic Unit | Time Frequency | Notes / Transformation |
|---|---|---|---|---|---|
| Residential electricity prices | Dependent variable | EIA861 | State - Virginia | Monthly - 2018-26<br><br>Yearly - 2018-26 | Removed other sectors and dates before 2018 and after 2025 |
| Inflation Rates | Control variable | Federal Reserve Bank of Minneapolis | National level - USA | Yearly - 2018-26 | Merge with population dataset |
| Electricity generation | Independent variable | SEDS | State - Virginia | Monthly - 2018-26 | Cleaning:<br>Added 0s to missing wind rows (no values because there were no turbines until late 2020).<br>Removed rows before 2018 and after 2025<br>Changed variable names<br>Removed dates in every column except the first |
| Residential electricity revenue, sales, and consumption | Dependent variable | EIA861 | State - Virginia | Monthly - 2018-26 | Removed dates before 2018 |
| Average retail electricity price | Dependent variable | EIA861 | State - Virginia | Captures 2024 year | Removed states that are not in Virginia |
| Electricity consumption | Independent variable | EIA | State - Virginia | Yearly - 2018-24 | Cleaning: removed 1960-2017 years |
| Data center location | Independent variable | ArcGIS | State - Virginia | Captures 2024-2026 when data centers are built | Removed unnecessary variables and variables with missing values in a significant number of |

| | | | | | |
|---|---|---|---|---|---|
| | | | | | rows. Removed notes in rows |
| Residential customers | Control variable | EIA861 | State - Virginia | Yearly - 2018-26 | Removed dates before 2018 and after 2025 |
| Data center electricity consumption | Independent variable | McKinsey & Company | State - Virginia | Yearly - 2018-23 | 2023-26 rows were assigned N/A |
| Natural Gas Prices | Independent variable | EIA | State - Virginia | Monthly - 2018-26 | Removed prices before 2018 and after 2025. Also removed rows with heading and notes. Changed date format to "yyyymm" |
| Henry Hub Natural Gas Prices | Independent | EIA | State - Virginia | Monthly - 2018-26 | Removed dates prior to 2018 and after 2025 |
| DOM Datacenter load | Independent variable | PJM Data Miner | State - Virginia | Monthly - 2018-2026 | Made yearly and monthly averages of hourly and daily frequency from the raw dataset |
| HDD | Independent variable | | State - Virginia | Monthly - 2018-2025 | Removed the column on stations used |
| CDD | Independent variable | | State - Virginia | Monthly - 2018-2025 | Removed the column on stations used |